Brock Biology of Microorganisms, 15e (Madigan et al.)
Chapter 32 Waterborne and Foodborne Bacterial and Viral Diseases

32.1 Multiple Choice Questions

1) The most important potential common source of infectious disease is
A) industrial waste.
B) arthropod reservoirs found near surface water supplies.
C) water.
D) human and animal wastes.
Answer: C
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 32.1

2) The degree of susceptibility a food has to microbial activity is determined by its
A) chemical characteristics.
B) physical characteristics.
C) water content.
D) chemical characteristics, physical characteristics, and water content.
Answer: D
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 32.6

3) Flour and sugar are classified as ________ foods.
A) highly perishable
B) semiperishable
C) nonperishable
D) selectively perishable
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.6

4) At approximately what temperature are most household freezers kept?
A) 4°C
B) 0°C
C) -20°C
D) -80°C
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.6

5) The rate of contaminant microbial growth during the exponential phase in food depends on
A) temperature.
B) the nutrient value of the food.
C) water content.
D) temperature, nutrient value, and water content.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.6

6) Psychrotolerant microorganisms can survive and grow at
A) refrigeration temperatures.
B) boiling temperatures.
C) extreme fluctuations in water availability.
D) high salt or sugar concentrations.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.6

7) The pH of most foods is
A) basic.
B) neutral or basic.
C) acidic.
D) neutral or acidic.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.6

8) The principle behind salt or sugar preservation is to
A) introduce a mechanical barrier to microbial invasion.
B) reduce water activity (a_w).
C) introduce a microbicide in anticipation of contaminating bacteria or fungi.
D) accomplish all of the described functions depending on the food being preserved.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.6

9) Pickling is a type of food preservation utilizing
A) weak acids.
B) basic substances.
C) lyophilization.
D) irradiation.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.6

10) The *Staphylococcus aureus* toxins are
A) neurotoxins.
B) endotoxins.
C) superantigen ectotoxins.
D) superantigen enterotoxins.
Answer: D
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.8

11) *Staphylococcus aureus* is a common causative agent of foodborne disease because it
A) grows on common foods.
B) is present in some humans that work in food processing.
C) produces several heat-stable enterotoxins.
D) grows on many foods, is present in some humans that work in food processing, and produces several heat-stable enterotoxins.
Answer: D
Bloom’s Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 32.8

12) Severe cases of *Staphylococcus aureus* food poisoning may require treatment
A) with powerful antibiotics.
B) for dehydration.
C) with cleansing enemas.
D) for dehydration, along with powerful antibiotics and cleansing enemas.
Answer: B
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.6

13) *Clostridium perfringens* food poisoning leads to diarrhea, because the
A) toxin causes the shedding of the intestinal lining.
B) permeability of the intestinal epithelium is altered by the toxin it produces.
C) disease causes such extreme thirst that the patient drinks more water than can be absorbed.
D) organism multiplies more rapidly than the immune system can handle.
Answer: B
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.9

14) *Clostridium botulinum* is a gram-________ rod that produces an ________.
A) negative / endotoxin
B) negative / exotoxin
C) positive endospore-forming / exotoxin
D) positive endospore-forming / endotoxin
Answer: C
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.9

15) Salmonellosis is most frequently caused by
A) *Salmonella enterica* serovar Typhi.
B) *Salmonella enterica* serovar Typhimurium.
C) *Salmonella enterica* serovar Enteriditis.
D) *Salmonella enterica* serovars Typhimurium and Enteriditis.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.10

16) Verotoxin can cause
A) hemorrhagic diarrhea.
B) liver failure.
C) brain damage.
D) hemorrhagic diarrhea, liver failure, and brain damage.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.11

17) ETEC strains produce one of two heat-________ diarrhea-producing ________.
A) stable / endotoxins
B) labile / endotoxins
C) stable / enterotoxins
D) labile / enterotoxins
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.11

18) The *Campylobacter* spp. are
A) aerobes.
B) anaerobes.
C) microaerophiles.
D) facultative anaerobes.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.12

19) *Listeria monocytogenes* is
A) acid-tolerant.
B) cold-tolerant.
C) salt-tolerant.
D) acid-, cold-, and salt-tolerant.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.13

20) Listeriosis is diagnosed from ________ cultures and treated with ________.
A) sputum / intravenous antibiotics
B) blood or spinal fluid / hydration therapy
C) sputum / hydration therapy
D) blood or spinal fluid / intravenous antibiotics
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.13

21) Raw food and food-handling equipment can be decontaminated with ________ to reduce the risk of listeriosis.
A) heat, but not radiation
B) radiation, but not heat
C) radiation and heat
D) neither radiation nor heat
Answer: B
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 32.13

22) The most common source of *Escherichia* spp. food poisoning is contaminated
A) meat.
B) fruits.
C) grains.
D) milk and milk products.
Answer: A
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 32.11

23) What organisms are the most likely pathogens to grow in food that has low water activity or solute-loaded?
A) gram-positive cocci
B) fungi
C) gram-negative bacilli
D) gram-positive bacilli
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.6

24) When solute is added to food products, the water activity (a_w)
A) increases.
B) decreases.
C) remains the same.
D) initially increases and then decreases.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.6

25) The water activity of food can be reduced by all of the following EXCEPT
A) physically removing water.
B) adding salt.
C) adding nitrites.
D) adding sugar.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.6

26) Irradiation of food uses
A) ionizing radiation.
B) a variety of chemical treatments.
C) desiccation.
D) non-ionizing radiation.
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.6

27) Which organisms are NOT involved in food fermentations?
A) lactic acid bacteria
B) sulfuric acid bacteria
C) acetic acid bacteria
D) propionic acid bacteria
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.6

28) The most common source of individual foodborne botulism outbreaks are due to consumption of
A) honey.
B) dairy products.
C) nonacid, home-canned vegetables.
D) egg and meat salads.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.9

29) The type of *Escherichia coli* that produces a verotoxin similar to the one produced by *Shigella dysenteriae* is ________ *E. coli*.
A) enterohemorrhagic
B) enterotoxigenic
C) enteropathogenic
D) enteroimmunogenic
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.11

30) Coliforms in a water sample indicate ________ contamination.
A) fecal
B) industrial or chemical
C) arthropod
D) All of the answers are correct.
Answer: A
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 32.2

31) Which U.S. government agency monitors water utility reports?
A) Environmental Protection Agency
B) Food and Drug Administration
C) Water Safety Board
D) Natural Resources Conservancy Commission
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.2

32) In the developed countries of the world, most intestinal infections are transmitted via
A) water.
B) person-to-person contact.
C) food.
D) hospital contamination.
Answer: C
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 32.1

33) The most common route of transmission of *Salmonella enterica typhi* worldwide is through
A) contaminated food.
B) contaminated water.
C) direct contact.
D) insect vectors.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.5

34) The presence of specific ________ signals that a given water source might be contaminated with pathogens.
A) exotoxins
B) chemical compounds
C) endotoxins
D) indicator microorganisms
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.2

35) In the United States, individuals become infected with *Vibrio cholerae* most commonly by
A) consuming contaminated shellfish.
B) drinking contaminated water.
C) eating contaminated vegetables.
D) person-to-person contact.
Answer: A
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.3

36) *Vibrio cholerae* primarily infects the
A) stomach.
B) small intestine.
C) large intestine.
D) rectum.
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.3

37) *Legionella pneumophila* is generally transmitted by
A) contaminated food.
B) contaminated drinking water.
C) contaminated water in coolers, pools, and domestic water systems.
D) person–to-person contact.
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.4

38) Which of the following is NOT characteristic of a coliform?
A) facultatively aerobic
B) gram-negative
C) rod shaped
D) resistant to chlorination
Answer: D
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.2

39) What is the medium used in the membrane filter (MF) procedure to differentiate fecal coliforms and *Escherichia coli*?
A) mannitol salt agar (MSA)
B) eosin-methylene blue agar (EMB)
C) triple sugar iron agar
D) blood agar
Answer: B
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.2

40) Which of the following bacterial pathogens is found in aquatic environments and is commonly present in air conditioning systems?
A) *Enterobacter aerogenes*
B) *Klebsiella pneumoniae*
C) *Legionella pneumophila*
D) *Vibrio cholerae*
Answer: C
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.4

32.2 True/False Questions

1) BOTH physical and chemical methods are used to treat and purify water.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.2

2) most wastewater treatment facilities employ methods designed to detect each pathogenic organism that may be present in a given sample.
Answer: FALSE
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 32.2

3) Local and state authorities can set standards for recreational water that are above or below the guidelines set by the United States government.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.1

4) Cholera can be transmitted by water or food.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.3

5) We are technically in the midst of a cholera pandemic.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.3

6) Infection with *Legionella pneumophila* is usually treated with intravenous erythromycin.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.4

7) Typhoid fever has been virtually (although not completely) eliminated from developed countries.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.5

8) Waterborne illness only occurs from exposure to contaminated drinking water, not from recreational use of contaminated water.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.1

9) Not all coliforms are pathogenic.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.2

10) The most important coliform for water quality testing is Escherichia coli because it is only found in the intestine.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.2

11) The critical step in making water safe to drink is chlorination.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.1

12) Norovirus is commonly transmitted by the fecal-oral route.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.5

13) The U.S. Environmental Protection Agency (EPA) regulates the concentration of *Escherichia coli* in public recreational waters as well as private swimming pools, spas, and hot tubs to minimize waterborne disease outbreaks.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.1

14) Legionellosis, caused by the bacterial respiratory pathogen *Legionella pneumophila*, has decreased in recent years due to advances in wastewater treatment procedures.
Answer: FALSE
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 32.4

15) Ideally, microbes should be eliminated from all foods.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.6

16) For much of the period of microbial growth in food, there is no visible or easily detectable change in food quality.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.6

17) Acidic foods generally need a higher canning temperature than do nonacidic foods.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.9

18) Properly canned foods typically do not contain living vegetative cells.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.6

19) Food poisoning generally results from the ingestion of food containing microbial toxins.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.7

20) Many food infection agents also cause waterborne diseases.
Answer: TRUE
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 32.7

21) With *Clostridium perfringens*, the enterotoxin is present in the endospores that germinate under anoxic conditions.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.9

22) To be certain the botulinum toxin is inactivated, the source must be heated for at least 2 minutes at 80°C.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.9

23) Salmonellosis can be caused by contaminated food or contaminated water.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.10

24) Antibiotic therapy is not very useful for treating typhoid fever.
Answer: FALSE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.5

25) Mass-processed ground meat is a particularly common source of infection with EHEC.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.11

26) Treatment of all pathogenic *E. coli* infections involves supportive therapy and, in some cases, may also include antimicrobial drugs.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.11

27) Invasive listeriosis is often treated with antibiotics such as penicillin.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.13

28) Gram-negative bacteria are responsible for causing over 60% of foodborne illness.
Answer: FALSE
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 32.7

29) *Campylobacter* infections are the most common foodborne infections in the United States.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.12

30) *Escherichia coli* is found naturally in the intestines of birds, humans, and animals.
Answer: TRUE
Bloom's Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.11

32.3   Essay Questions

1) Explain why coliforms are used to detect water contamination rather than directly quantifying individual pathogens such as *Vibrio cholerae* or *Salmonella spp*.
Answer: Coliforms, such as *Escherichia coli*, are detected in water samples to indicate the potential contamination of human or other animal feces. Because coliforms are shed in animal fecal matter, higher E. coli concentrations in water also suggest high numbers of other animal-related fecal borne pathogens present. The rapid growth rate of *E. coli* provides quick results.
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 32.2

2) Explain how the membrane filter (MF) procedure is used.
Answer: The MF procedure is used to provide information on the concentration of coliforms and *E. coli*. The procedure involves filtering a water sample such that the microbial cells are trapped onto the filter, and the filter is placed onto a differential and selective growth medium such as a eosin-methylene blue (EMB) agar plate. The agar plate is incubated and colony-forming units are directly counted.
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 32.2

3) There were numerous cases throughout the United States of patients having bloody diarrhea all within a week of each other, many of which were hospitalized with severe cases. Although the people spanned from California to New York, they all had attended a high school graduation party. Epidemiologists determined the common food factor was cheese curds homemade by a farmer. The home where the cheese curds were made was searched, and the cheese curds and milk used to produce them were taken to the local government lab. Among other microbes, *Campylobacter jejuni* was isolated from many of the cheese curds as well as the farmer′s bulk tank milk. What should be done for those infected? What can be done to prevent this from happening again?
Answer: Antibiotics are usually not given with *C. jejuni* infections, because the infections are self-limiting. However, in more prolonged infections where the symptoms may be more severe, antibiotics can be used. Since this is a more severe case with a prolonged time period, antibiotics could be given to those hospitalized (e.g., erythromycin or a fluoroquinolone), which will shorten the duration of symptoms if they are given early in the illness. Pasteurizing the milk used to make the cheese curds should kill *C. jejuni* and prevent reoccurrence.
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 32.12

4) What makes *Listeria monocytogenes* an especially difficult-to-prevent foodborne pathogen? Once in the human body, how does it evade the immune system?
Answer: *L. monocytogenes* grows well at refrigerator temperatures of 4°C, because it is psychrotolerant, so even properly stored foods are suitable for growth of the pathogen. Once phagolysosomes of the pathogen are made in response to its presence in the gastrointestinal tract, they are lysed by a listeriolysin produced by *L. monocytogenes*. The pathogen propagates in the cytoplasm and pushes itself into the cytoplasmic membrane where it forms filopod protrusions. Other surrounding cells ingest the protruded filopods, and the entire proliferation cycle can continue while evading antibodies, complement, and neutrophils.
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 32.13

5) Why are enteric bacteria such as *Escherichia coli* common meat contaminants?
Answer: *E. coli* and other enterics naturally inhabit other animals′ organs such as the intestines, where they generally do not cause harm. Improper handling when such infected organs are removed can result in accidental contamination of meat. Infected ground meats are especially of concern, because the meat must then be fully cooked to kill the pathogens.
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 32.11

6) How does *Clostridium perfringens* cause food poisoning? How is this process related to the metabolism and special structures of *C. perfringens*?
Answer: *C. perfringens* most commonly infects meats, and if they are not heated to a high enough temperature, endospores will be produced by the pathogen. Endospores will germinate under anoxic conditions, so canning a *C. perfringens* endospore-infected food will induce germination of the pathogen. When food that contains *C. perfringens* is ingested, the cells will again sporulate under anoxic conditions in the human intestine as well as produce enterotoxins. The epithelial lining of the intestine is permeabilized by the enterotoxin, which leads to intestinal cramps, diarrhea, and nausea. The onset of symptoms usually begins around 7-15 hours after ingestion, and the disease as a whole is normally cleared within 24 hours in an otherwise healthy individual.
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 32.9

7) Why is *Staphylococcus aureus* often described as an international picnic pest?
Answer: *S. aureus* is a member of at least 20-30% of people′s normal flora, so it is easily transmitted to foods, where it is capable of growth on several different foods. The pathogen grows very well at room temperature or slightly warm outdoor temperatures. Staphylococcal food poisoning is not caused by the bacterium but by the enterotoxin it produces during growth, which means the food could have been heated, and the heat-tolerant enterotoxins could have survived. Mayonnaise-based salads as well as custard- and cream-filled desserts are among the most common foods associated with staphylococcal food poisoning.
Bloom’s Taxonomy: 1-2: Remembering/Understanding
Chapter Section: 32.8

8) What is the difference between food poisoning and food infection? Are antibiotics useful in either of them?
Answer: Food poisoning, also called food intoxication, results from the ingestion of foods containing preformed microbial toxins. Foodborne infections result from ingestion of pathogen-contaminated food. Antibiotics are not normally used to treat food poisoning, because the toxin is causing the disease. Food infections can be treated with antibiotics to kill the pathogen(s) more quickly but are not always used if the immune system is expected to be strong enough to rid the body of the infection on its own.
Bloom’s Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 32.7

9) As a visitor to a country in which cholera is an endemic disease, what specific steps would you take to reduce your risk of cholera exposure? Will these precautions also prevent you from contracting other waterborne diseases? If so, which ones? Identify waterborne diseases for which your precautions may not prevent infection.
Answer: To avoid contracting cholera, travelers should wash their hands and/or use a ethanol-based hand sanitizer regularly, as well as drink only bottled water or other bottled beverages. Ice, raw food (even raw vegetables or fruit), and raw or undercooked fish and shellfish should be avoided as well. These precautions will help prevent other waterborne diseases such as typhoid fever and norovirus, but will NOT prevent legionellosis.
Bloom's Taxonomy: 5-6: Evaluating/Creating
Chapter Section: 32.3

10) Identify and define the categories of food perishability. Compare the perishability of sugar, milk, and hard cheeses (such as parmesan).
Answer: Foods are classified as perishable, semiperishable, and nonperishable. Perishable foods are generally fresh and have high water activity. Semiperishable foods have a lower water activity or have been preserved in some way. Nonperishable foods have a very low water activity and can be stored for long periods of time without spoilage. Milk is considered a perishable food with a high water activity and nutrient content; it must be refrigerated to avoid spoilage. Hard cheeses would be semiperishable because the added acid, salt and dehydration that occurs during cheese production. Granulated sugar has a very low water activity and is nonperishable.
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 32.6

11) Why are antibiotics of little value by themselves in the treatment of cholera?
Answer: The disease cholera is not caused by the bacterium *Vibrio cholera* itself but rather the enterotoxin that it produces. While antibiotic treatment will halt enterotoxin production, it does not inactivate the enterotoxins already synthesized and therefore does not immediately treat the disease.
Bloom's Taxonomy: 3-4: Applying/Analyzing
Chapter Section: 32.3